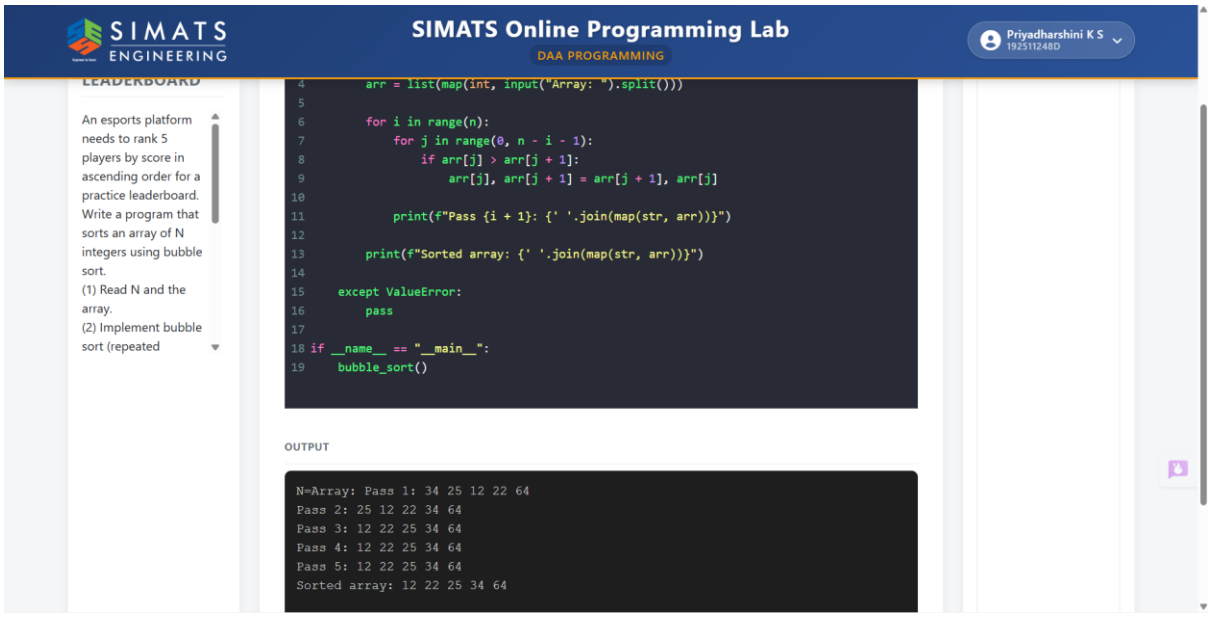
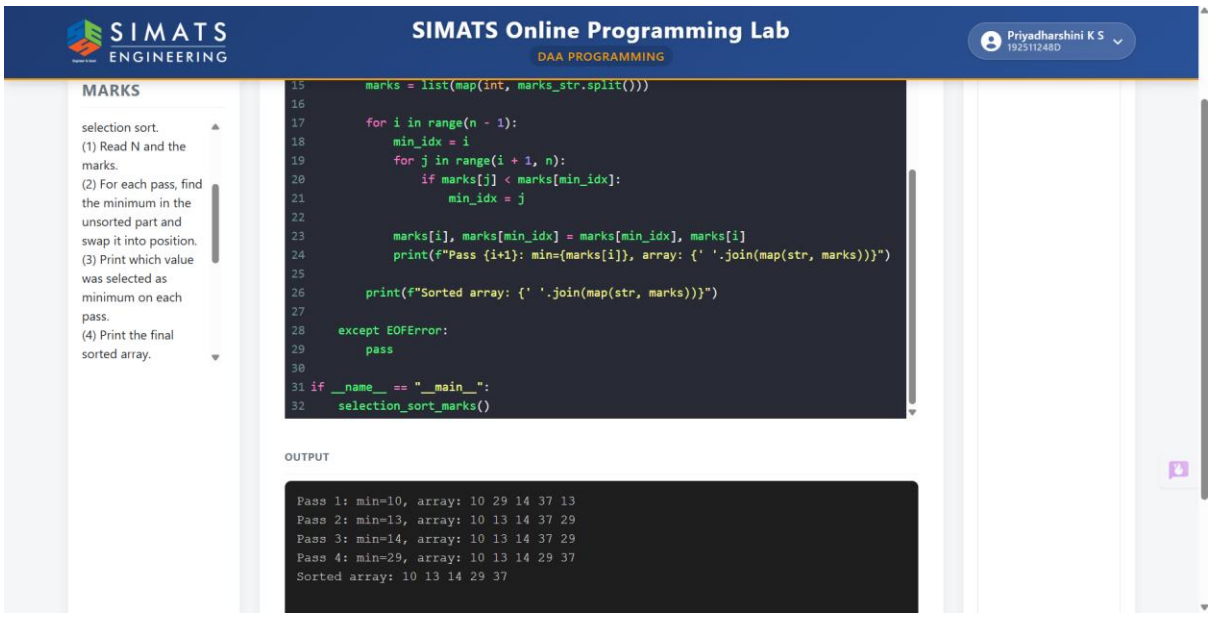
BRUTE FORCE SORTING





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INSERTION SORT QUEUE

array of N heights using insertion sort.
(1) Read N and the heights in arrival order.
(2) For each new element, shift larger elements right and insert it in place.
(3) Print the line after each insertion.
(4) Print the final sorted line.

```
try:
    n_input = input("N=").strip()
    if not n_input:
        return
    n = int(n_input)

    heights_input = input("Heights: ").strip()
    if not heights_input:
        return
    original_heights = list(map(int, heights_input.split()))

    for i in range(1, n):
        key = original_heights[i]
        temp_list = original_heights[:i+1]
        for k in range(1, len(temp_list)):
            val = temp_list[k]
            m = k - 1
            while m >= 0 and temp_list[m] > val:
                temp_list[m + 1] = temp_list[m]
```

OUTPUT

```
N=Heights: After inserting 142: 142 150 160 138 155
After inserting 160: 142 150 160 138 155
After inserting 138: 138 142 150 160 155
After inserting 155: 138 142 150 155 160
Final line: 138 142 150 155 160
```

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Bubble Sort Swap C

BUBBLE SORT SWAP COUNT

array using bubble sort while counting the total number of swaps performed.
(1) Read N and the array.
(2) Perform bubble sort, incrementing a swap counter on every swap.
(3) Print the sorted array and total swap count.

```
            arr = list(map(int, line2.split()))

        swaps = 0
        # Bubble Sort Algorithm
        for i in range(n):
            for j in range(0, n - i - 1):
                if arr[j] > arr[j + 1]:
                    # Swap the elements
                    arr[j], arr[j + 1] = arr[j + 1], arr[j]
                    swaps += 1

        print(f"Sorted array: {' '.join(map(str, arr))}")
        print(f"Total swaps: {swaps}")

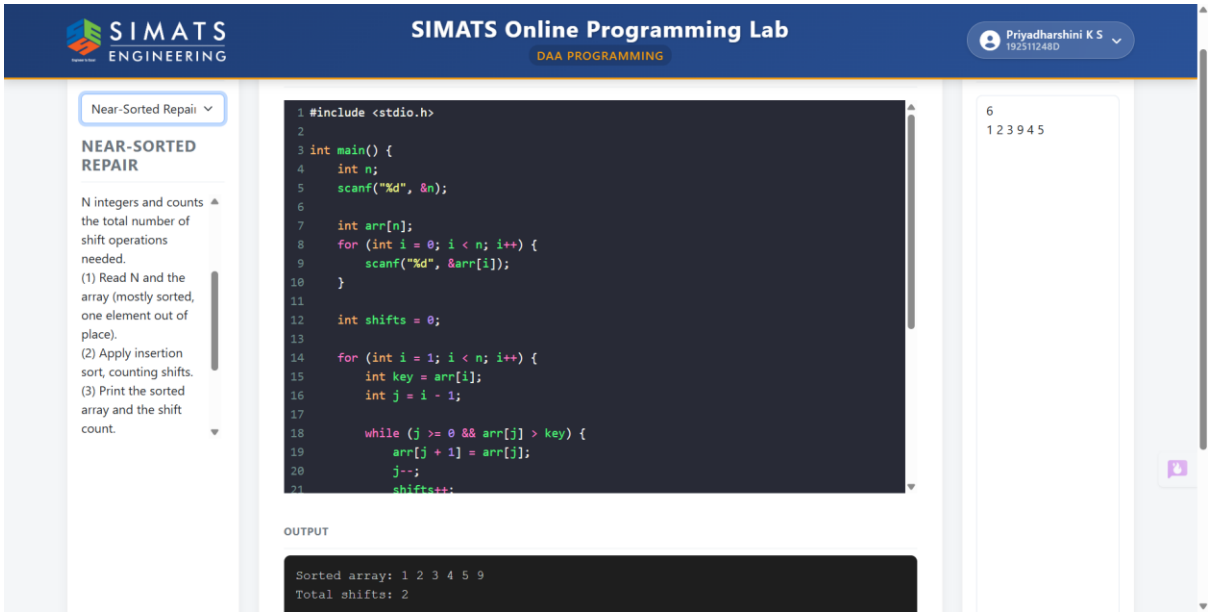
    except EOFError:
        pass
    except Exception as e:
        pass

if __name__ == "__main__":
    bubble_sort_with_swap_count()
```

6
6 5 4 3 2 1

OUTPUT

```
Sorted array: 1 2 3 4 5 6
Total swaps: 15
```


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Near-Sorted Repair
NEAR-SORTED REPAIR
N integers and counts the total number of shift operations needed.
(1) Read N and the array (mostly sorted, one element out of place).
(2) Apply insertion sort, counting shifts.
(3) Print the sorted array and the shift count.
#include <stdio.h>
int main() {
int n;
scanf("%d", &n);
int arr[n];
for (int i = 0; i < n; i++) {
scanf("%d", &arr[i]);
}
int shifts = 0;
for (int i = 1; i < n; i++) {
int key = arr[i];
int j = i - 1;
while (j >= 0 && arr[j] > key) {
arr[j + 1] = arr[j];
j--;
shifts++;
6
1 2 3 9 4 5
OUTPUT
Sorted array: 1 2 3 4 5 9
Total shifts: 2